

Large language models in human-robot collaboration: A systematic review, trends, and challenges

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ABSTRACT

The integration of large language models (LLMs) into human-robot collaboration (HRC) represents a paradigm shift toward cognitive manufacturing under Industry 5.0. This systematic review analyzes 1278 publications to elucidate the enabling mechanisms, application hierarchies, and persistent challenges of LLM-enhanced HRC. We identify three agent paradigms, each tailored to structured, interactive, and sensor-rich scenarios, respectively. Furthermore, we propose a layered training framework combining domain-adaptive pre-training and scenario-specific fine-tuning to bridge general LLMs capabilities with industrial demands. Our findings reveal that LLMs significantly advance HRC in environmental perception, long-horizon task planning, and embodied interaction, yet face critical challenges in dynamic adaptability, multimodal fusion, and value alignment. This study provides a structured reference for developing robust, interpretable, and human-centric collaborative systems, outlining five key research directions to guide future advancements in cognitive manufacturing.

Introduction

The swift advancement of artificial intelligence is reshaping the paradigms governing human-robot collaboration (HRC) within the manufacturing sphere. While traditional HRC have proven effective in structured environments, they exhibit considerable limitations when faced with dynamic shop-floor conditions, open-ended tasks, and the necessity for naturalistic human-robot collaboration [1–3]. These challenges underscore the urgent need for intelligent systems adept at real-time environmental interpretation, adaptive planning, and resilient execution amidst uncertainty. The emergence of large language models (LLMs), equipped with unparalleled abilities in semantic understanding, contextual reasoning, and task decomposition, presents opportunity to overcome these challenges [4–6]. Trained on extensive multimodal corpora, LLMs showcase astonishing emergent abilities including intricate logical planning [7], cross-modal understanding [8], and embodied instruction-following [9], all of which resonate with the cognitive demands inherent to industrial HRC.

Recent advancements elucidate remarkable strides in environmental adaptability through the integration of large language models into HRC systems [10,11]. These sophisticated systems enhance performance

across three pivotal dimensions, namely, intent interpretation in unstructured contexts via nuanced natural language processing techniques [12,13], dynamic task planning driven by real-time reasoning amidst evolving constraints [14,15], and resilient execution facilitated by probabilistic adaptations to environmental uncertainties [16,17]. This confluence of technologies heralds significant progress toward cognitive manufacturing paradigms wherein humans and robots engage in seamless collaboration [18]. Nonetheless, fundamental challenges endure in the practical application of these innovations. First and foremost, adapting to dynamic environments demands groundbreaking advancements in real-time perception coupled with replanning strategies capable of addressing physical uncertainties such as object displacement or task evolution [19]. Secondly, multimodal fusion bottlenecks arise from the heterogeneous attributes inherent to sensory data, encompassing vision, language, and haptics, and the intricate complexities associated with spatiotemporal alignment. Lastly, ensuring cognitive-value alignment necessitates verifiable mechanisms that safeguard compliance with human intention while upholding safety protocols and ethical norms, prerequisites essential for reliable deployment within diverse operational landscapes.

Despite the burgeoning scholarly interest, prevailing surveys on the integration of LLMs and robotics reveal significant deficiencies [20–22].

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Nomenclature		LLMs	Large language models
AI	Artificial intelligence	MAE	Mean absolute error
AR	Augmented reality	MTTR	Mean time to recovery
ASR	Adversarial success rate	MFA	Multimodal fusion agents
AEA	Autonomous execution agents	MR	Mixed reality
BCI	Brain-computer interfaces	RLFT	Reinforcement learning fine-tuning
CA	Cognitive alignment	MCA	Multimodal contextual awareness
DAP	Domain-adaptive pre-training	NG	Navigation guidance
EQA	Embodied question answering	PT	Preliminary training
FPFT	Full-parameter fine-tuning	PEFT	Parameter efficient fine-tuning
PEFT	Parameter efficient fine-tuning	PFT	Prompt fine-tuning
GCP	General corpus pre-training	RP	Replanning
HRC	Human-robot collaboration	RLHF	Reinforcement learning from human feedback
HAI	Human-AI interaction	RLFT	Reinforcement learning fine-tuning
ICA	Interactive coordination agents	RR	Representational reasoning
ITA	Interactive agents	RQs	Research questions
LoRA	Low-rank adaptation	SLR	Systematic literature review
LHM	Long-horizon manipulation	SAP	Safety-aligned pre-training
LIT	Language instructions	VLT	Vision language tracking
		VLMS	Visual language models

The extant literature predominantly centers around technical architectures or discrete application instances, lacking analysis of embodied reasoning frameworks within manufacturing environments. Moreover, there is an alarming scarcity of rigorous review concerning the evolving trends in implementation and the multifaceted deployment challenges, particularly with regard to industrial constraints. This oversight becomes pertinent in light of the explosive advancement in embodied LLMs research, which necessitates a synthesis to inform and steer future developments effectively.

To bridge these existing gaps, this work undertakes a systematic review with the objective of addressing the subsequent research questions (RQs).

RQ1. : What are the uniqueness and enabling technologies of LLMs in human-robot collaboration?

RQ2. : What are the prominent research themes and presentations regarding HRC informed by LLMs?

RQ3. : What potential challenges and emerging trends in HRC should be anticipated with respect to the integration of LLMs?

Through a detailed analysis of references from 2015 to 2025, this work has made three key contributions: 1) expounding the uniqueness and enabling technologies of LLMs to provide conditions for repeatable experiments; 2) conducting an in-depth analysis of the technical details of HRC scenarios to achieve a universal interpretation; 3) distilling the cutting-edge directions and actual obstacles of HRC to provide references for subsequent research.

The subsequent sections of this work are delineated as follows (in Fig. 1). Section 2 delineates the methodology employed in the review, along with the analysis for bibliometric. Section 3 presents the enabling mechanisms of LLMs for HRC. Section 4 delves into an intricate examination of the studies identified, providing insights and key findings. Section 5 synthesizes both technical and challenges while articulating potential avenues for future research. Section 6 serves as a summary of the work.

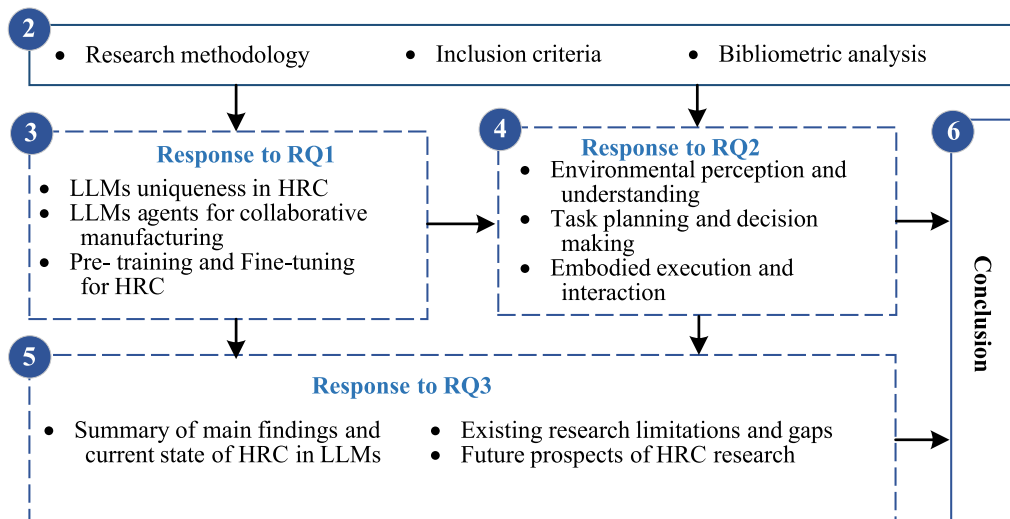


Fig. 1. An overview of this review.

Bibliometric analysis

Retrieval and screening

The systematic literature review (SLR) method represents a replicable approach to literature analysis, aimed at gathering, assessing, and synthesizing all pertinent research on a designated topic through a series of procedures [23–25]. This process minimizes bias while yielding trustworthy conclusions. It is undeniable that the SLR is shaped by the data sources utilized. The Web of Science and Scopus stand out as pre-eminent databases rich in academic resources [26,27]. In this work, we employed the SLR to undertake an extensive multi-dimensional textual examination of relevant works pertaining to LLMs in HRC. We executed a search across these three esteemed databases and selected high-caliber research papers for analysis.

The research was performed in Dec 2025, using the following publication databases: Web of Science and Scopus. The search phrase utilized was: “human robot language model” or “human robot LLM” or “human robot VLM” or “human machine language model” or “human machine LLM”. This set of keywords was searched in the title, abstract and keywords records of the journal articles and conference proceedings written in English, from 2015 to 2025. The selection criteria for these terms are predicated on the notion that the document must encompass extensive models and facilitate HRC, acknowledging that this process is fundamentally reliant upon the synergy between human and robotic counterparts. Following a myriad of attempts employing diverse keyword combinations within the database, we inputted this array of search parameters. Consequently, a total of 1998 results emerged from Web of Science (comprising 726 papers and 1066 proceeding Papers). From Scopus, an aggregate of 2366 results were obtained (including 1180 papers and 1351 conference proceedings). To screen and extract valuable insights from the targeted literature, this study employed a three-tiered fine screening process, as illustrated in Fig. 2. Initially, these two databases were harmonized and duplicates eliminated, resulting in a transition from an original pool of 4364 papers to a consolidated set comprising 2092 works. Furthermore, publications that engaged with this topic yet failed to present empirical experimental research were excluded from consideration. We peruse the title and abstract of each selected paper to ascertain its incorporation of pertinent keywords pertaining to the topic, resulting in 1665 entries. Ultimately, through a reading and rigorous analysis of the entire text, all the documents have been contextualized. This endeavor has evaluated results, methods and processes, resulting in 1278 entries. Notably, within these works, the total number of instances may exceed that indicated in the individual papers themselves, as certain studies present multiple manifestations of identical features.

Results analysis

The annual number of published papers and citations in this field has

exhibited a significant accelerating growth trend over the past decade in Fig. 3, confirming the rapid increase in its academic influence and research popularity. According to the data, from 2015 to 2021, the volume of published articles demonstrated a relatively steady growth trajectory, averaging approximately 50–70 publications per year. In parallel, citation counts increased from just over ten annually to more than one thousand. However, since 2022, both metrics have entered a phase of steep growth. By 2025, preliminary data indicated that there were already 404 published articles and 7170 citations recorded. It is anticipated that these annual figures will continue to reach new heights. This inflection point in growth closely aligns with breakthroughs and widespread applications of generative artificial intelligence (AI), particularly marked by the release of ChatGPT-3.5 at the end of 2022, which stimulated research interest and technological exploration within the global academic community across related domains. Notably, the rate of citation growth surpasses that of published articles considerably. This observation reflects not only an increase in research outputs within this field but also an expansion in the influence and dissemination scope of its academic contributions. In summary, these data indicate that this field has entered an active development stage characterized by concentrated output, a pattern consistent with typical technology-driven research trajectories.

The publication of research achievements in this domain exhibits a concentrated characteristic, with the evolution of publishing platforms aligned with the overall increase in the field's popularity. As illustrated in Table 1, journals and flagship conferences under IEEE series an unequivocal dominant position. The forms of academic publishing exhibit a diversified and complementary pattern. Moreover, proceedings from computer science conferences also encompass numerous studies, highlighting the significant intersection between this field and disciplines such as artificial intelligence and computer science. Furthermore, open-access journals uphold stable publication records, serving as a crucial outlet for related applied research. It is noteworthy that there are substantial variations in research activity levels over time, further corroborating the accelerated emergence of hotspots within this field. While most sources maintained relatively stable article publication rates between 2021 and 2023, there was a marked increase observed during 2024–2025, particularly among journals and conferences pertaining to robotics and manufacturing systems. This shift is linked to advancements in generative AI that propel developments in HRC, indicating that HRC research is penetrating high-value application scenarios within intelligent manufacturing. Highly cited literature clearly outlines the core context and cutting-edge trends of knowledge evolution in this field. In Table 2, the vast majority of highly cited literature has seen an exponential increase in citations since 2023. This is highly synchronized with the technological wave of generative AI, indicating that breakthrough technologies are the key driving force behind the creation of core literature. The top-ranked literature mainly focuses on the social impact of large language models, evaluation methods, human-robot interaction and robot learning fields, forming a clear research focus

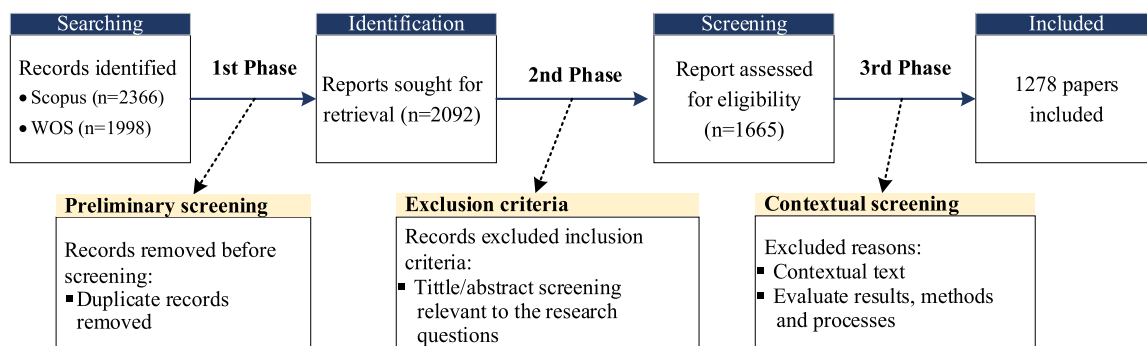


Fig. 2. Procedure for the selection of review articles.

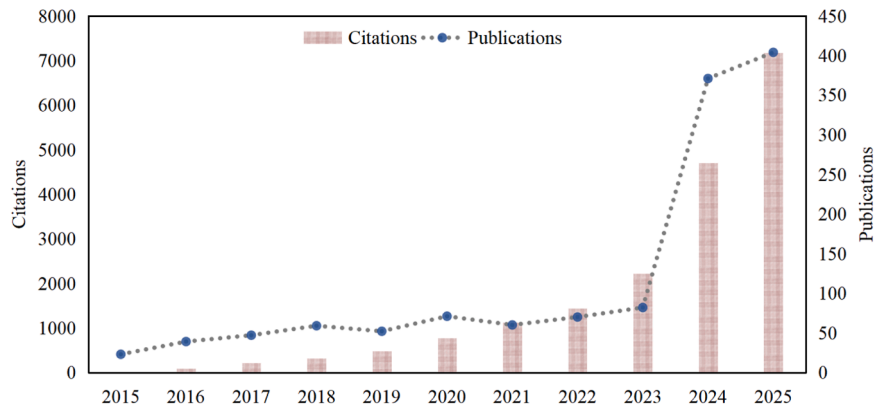


Fig. 3. Annual number of publications and citations.

Table 1

Top publishing sources for the topic research by years of publication.

Source	Years of publication					Total
	2021	2022	2023	2024	2025	
IEEE Robotics and Automation Letters	4	3	3	23	32	65
Lecture Notes in Artificial Intelligence	3	3	3	8	28	45
IEEE International Conference on Human Robot Interaction	3	3	2	4	32	44
Lecture Notes in Computer Science	2	6	2	11	21	42
IEEE International Conference on Intelligent Robots and Systems	2	2	9	29	0	42
Proceedings of Machine Learning Research	4	3	12	22	0	41
IEEE Access	2	2	5	4	24	37
IEEE International Conference on Robots and Automationica	0	0	6	21	0	27
Applied Sciences Basel	5	3	1	7	5	21
Robotics and Computer Integrated Manufacturing	1	0	0	3	15	19
Sensors	4	1	3	3	6	17
Electronics	1	0	0	4	12	17
International Journal of Social Robotics	5	2	0	3	6	16
Journal of Manufacturing Systems	0	0	1	2	10	13
Robotics and Autonomous Systems	1	1	2	0	3	7
IEEE Transactions on Automation Science and Engineering	1	0	0	1	4	6
IEEE CAA Journal of Automatica Sinica	0	1	1	1	2	5
Engineering Applications of Artificial Intelligence	0	0	0	2	3	5
CIRP Annals Manufacturing Technology	0	0	0	3	1	4
Autonomous Robots	0	1	1	0	2	4

cluster. These highly cited literatures not only include systematic reviews and risk discussions of new technologies, but also cover interdisciplinary application explorations and fundamental research methods.

In the keyword co-occurrence network generated by CiteSpace, high-frequency and central nodes clearly outline the research focus and evolution frontier of this field. The core keyword cluster is centered on HRC and closely related to the two enabling technologies of artificial intelligence and machine learning, as shown in Fig. 4, indicating that the

research paradigm in this field has deeply shifted towards intelligence. It is worth noting that large language models and natural language processing, as emerging key nodes, have significant strength and are strongly connected with core clusters. This indicates that the research frontier is rapidly shifting from traditional instructions and control to a more intuitive and efficient HRC and task collaboration paradigm based on natural language. Meanwhile, the emergence of emotion recognition as an important sub-node reveals the current research focus on achieving deep and contextualized human-robot understanding and empathy. Overall, the keyword network indicates that this field is integrating cutting-edge AI technologies, jointly promoting the development of intelligent HRC towards a new stage of naturalization, anthropomorphism and emotionalization.

The construction of the author co-occurrence knowledge graph reveals the intricate interrelationships between authors and co-authors within the field. We extracted the knowledge graph of author co-occurrence from the author data analysis, as shown in Fig. 5. This graph has 290 nodes and 324 edges, indicating a strong interrelationship among the authors. This indicates that multiple research teams have initially taken shape, and some teams have also established certain cooperative networks among themselves. One of the outstanding team numbers at the node is Taniguchi, Nagai, and Cangelosi. The focus of another team revolves around Thomas, Roy and Walter. In addition, the team numbers of Zheng, Wang and Fan are emerging talents, and their posts have increased sharply in the recently years. Overall, the author collaboration network reflects that this field is in a stage of expanding from being led by individual core teams to a more diverse and cross-disciplinary collaboration ecosystem.

The enabling mechanisms of LLMs for HRC

To methodically deconstruct the role of LLMs in HRC, this chapter establishes a layered analytical framework for the underlying theoretical knowledge. Moving beyond a mere exposition of generic LLMs technologies, our classification is deliberately structured along three pivotal dimensions, the LLMs agent paradigm, pre-training and fine-tuning strategies. This structure is informed by a progressive principle that transitions from foundational capabilities to functional specialization [48], aiming to answer a central question: What constitutes an LLMs for HRC, and how is it technically constructed? Consequently, we illuminate the distinct advantages of various agent paradigms in diverse HRC scenarios. We also elucidate how pre-training and fine-tuning methodologies act as the critical bridge, facilitating the seamless transfer of general LLMs into specific, reliable, and efficient HRC contexts. In Table 3, we conducted a detailed examination of the agents, pre-training and fine-tuning of the 65 key literature. Through this systematic analysis, which moves from the agent's role to its internal mechanics and finally to its domain adaptation, this chapter provides a cohesive

Table 2

Most-cited publications in the topic research.

Ref.	Title	Source	WOS Times cited (By Dec 2025)						Scopus cited
			2021	2022	2023	2024	2025	Total	
[28]	A brief overview of ChatGPT: the history, status Quo and potential future development	IEEE-CAA Journal of Automatic Sinica	0	0	29	363	402	797	1113
[29]	Adoption of AI-based chatbots for hospitality and tourism	International Journal of Contemporary Hospitality Management	28	60	115	160	166	530	722
[30]	Why Johnny can't prompt: how non-AI experts try (and fail) to design LLM prompts	CHI conference on Human Factors in Computing Systems	0	0	21	224	230	476	649
[31]	Safe learning in robotics: from learning-based control to safe reinforcement learning	Annual Review of Control Robotics and Autonomous Systems	0	19	116	184	116	437	578
[32]	From human writing to artificial intelligence generated text: examining the prospects and potential threats of ChatGPT in academic writing	Biology of Sport	0	0	44	121	152	320	497
[33]	Taxonomy of risks posed by language models	Conference on Fairness Accountability and Transparency	0	1	62	126	125	315	467
[34]	A review of verbal and non-verbal human-robot interactive communication	Robotics and Autonomous Systems	48	39	30	27	21	300	369
[35]	The robots are coming: exploring the implications of OpenAI codex on introductory programming	24th Australasian Computing Education Conference	0	2	58	132	58	252	353
[36]	REVERIE: Remote embodied visual referring expression in real indoor environments	IEEE/CVF Conference on Computer Vision and Pattern Recognition	20	32	48	56	50	212	285
[37]	Language2Pose: Natural language grounded pose forecasting	7th International Conference on 3D Vision	7	25	37	61	43	176	244
[38]	Systematic review and meta-analysis of AI-based conversational agents for promoting mental health and well-being	NPJ Digital Medicine	0	0	0	38	111	150	183
[39]	Facilitating human-robot collaborative tasks by teaching-learning-collaboration from human demonstrations	IEEE Transactions on Automation Science and Engineering	23	23	19	17	18	142	143
[40]	A review of eye gaze in virtual agents, social robotics and HCI: behaviour generation, user interaction and perception	Computer Graphics Forum	11	15	14	25	14	140	171
[41]	Can large language models be an alternative to human evaluation?	Annual Meeting of the the Association for Computational Linguistics	0	0	5	71	58	135	285
[42]	ChatGPT and large language model (LLM) chatbots: The current state of acceptability and a proposal for guidelines on utilization in academic medicine	Journal of Pediatric Urology	0	0	10	49	65	126	166
[43]	Wearable sensor-based human activity recognition with transformer model	Sensors	0	10	33	45	35	123	167
[44]	Video description: A survey of methods, datasets, and evaluation metrics	ACM Computing Surveys	35	25	26	15	15	123	152
[45]	Tell me Dave: Context-sensitive grounding of natural language to manipulation instructions	International Journal of Robotics Research	14	16	10	18	6	109	151
[46]	The body talks: Sensorimotor communication and its brain and kinematic signatures	Physics of Life Reviews	19	16	19	13	12	106	107
[47]	Survey of emotions in human-robot interactions: perspectives from robotic psychology on 20 years of research	International Journal of Social Robotics	0	13	23	38	28	102	117

foundation for answering RQ1.

Large language model uniqueness in HRC

The application of LLMs in HRC within the manufacturing industry presents significant uniqueness and challenges that are distinct from those observed in other sectors [49,50]. This uniqueness is rooted in the manufacturing industry's ultimate pursuit of precision, reliability, safety, and efficiency [51,52]. As a result, the technical pathways and application paradigms for LLMs in HRC settings exhibit unique domain characteristics. To analyze the distinctive impact of LLMs on the manufacturing sector, we selected service robots and human-AI interaction (HAI) as comparative fields. Service robots epitomize embodied physical interactions closely related to HRC within manufacturing, thereby facilitating the identification of core differences among similar types of robots. In contrast, HAI represents a broader interaction paradigm that highlights specific particularities pertinent to manufacturing concerning tasks, environments, and constraints.

In terms of task characteristics, the manufacturing HRC imposes the most stringent requirements on LLMs. Tasks within industrial manufacturing contexts typically adhere to well-defined process standards and operational norms [53]. The instructions generated by LLMs must be capable of being translated into high-precision physical actions,

with outputs that comply with strict process parameters and physical constraints. In contrast, the boundaries of tasks in the domain of service robots are comparatively ambiguous [54]. For instance, directives such as "assisting the elderly in their daily lives" permit various implementation methods and allow for a relatively wide margin of error. In the field of HAI, tasks often remain confined to information exchange and do not entail complex processes involving physical execution. From the perspective of environmental characteristics, the HRC in the manufacturing industry embodies a unique complexity where high structuring and dynamic uncertainty coexist [55]. The manufacturing environment is founded upon rigorous process systems and equipment standards while simultaneously requiring adaptation to unforeseen circumstances such as material deviations and equipment malfunctions. This characteristic necessitates that LLMs not only master structured manufacturing knowledge but also develop capabilities for uncertain reasoning within a deterministic framework. In contrast, the working environment for service robots is typically entirely unstructured, demanding solutions to common open-ended problems. Meanwhile, HAI environments are relatively simplistic, primarily confined to an information space. In terms of interaction modes, HRC exhibits distinct characteristics. In industrial scenarios, HAI prioritizes task completion as its primary objective, with a strong emphasis on efficiency, accuracy, and security. The interactive content is both highly specialized,

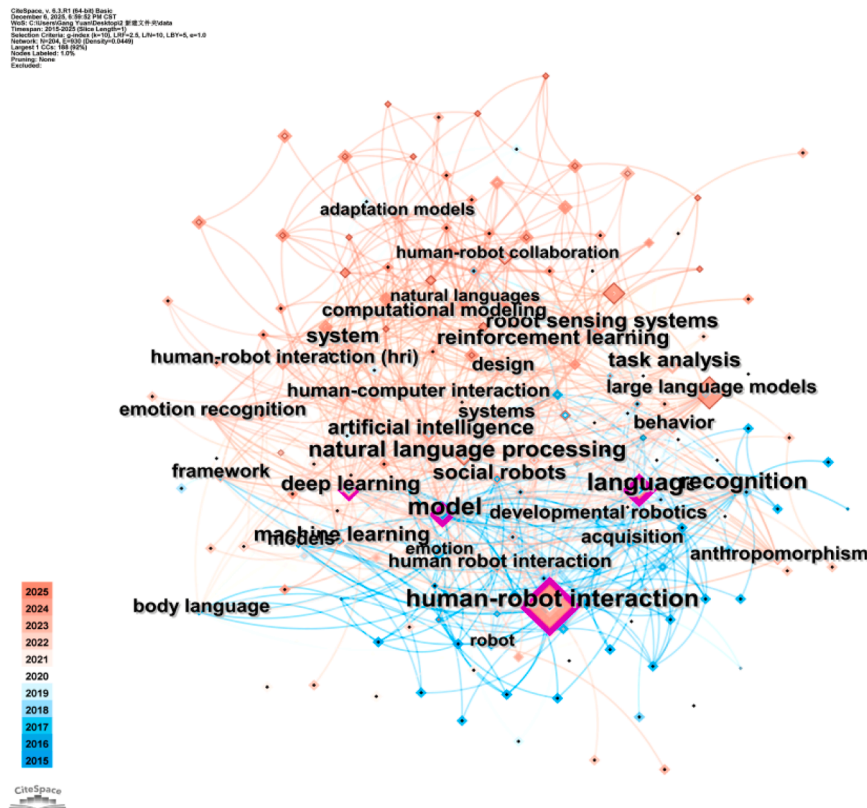


Fig. 4. Keywords co-occurrence knowledge graph.

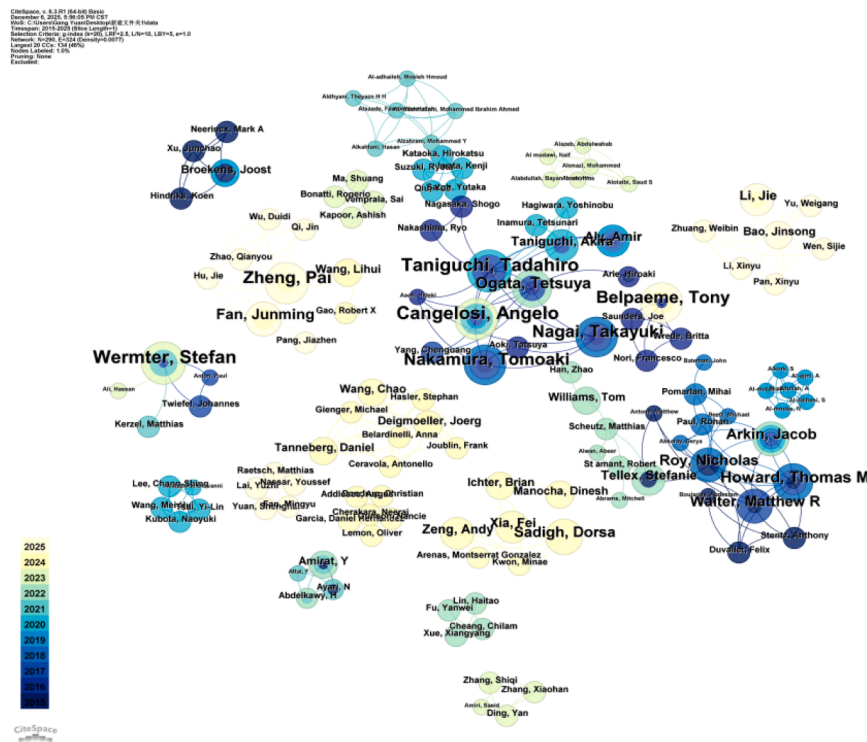


Fig. 5. Authors co-occurrence knowledge graph.

involving numerous domain-specific terms and standards, and inherently multimodal [56]. It demands the seamless integration of multiple sensory inputs such as vision and force perception, all under stringent

real-time constraints. The design of service robot interactions is often driven by considerations of user experience and emotional factors, which consequently allows for more personalized and flexible

Table 3
Key literature review for agents, pre-training, fine-tuning.

Reference	Year	Intelligent agent			Pre-training			Fine-tuning			
		AEA	ICA	MFA	GCP	DAP	SAP	FPFT	PEFT	PFT	RLFT
[49] Lou et al.	2025	✓	✓	✓	✓		✓			✓	
[50] Wang et al.	2026	✓	✓		✓					✓	
[51] Ma et al.	2026	✓		✓	✓	✓	✓		✓	✓	
[52] Chen et al.	2026	✓	✓	✓	✓		✓			✓	
[53] Tasneem et al.	2026	✓			✓					✓	
[54] Wang et al.	2025		✓		✓		✓	✓		✓	✓
[55] Yin et al.	2025		✓	✓	✓	✓		✓		✓	✓
[56] Liu et al.	2026	✓	✓	✓	✓		✓	✓			
[57] Ding et al.	2026	✓	✓	✓	✓	✓		✓	✓		
[58] Liu et al.	2026	✓	✓	✓	✓						
[59] Wang et al.	2026	✓	✓		✓	✓				✓	
[60] Wang et al.	2026	✓	✓		✓					✓	✓
[61] Yu et al.	2025	✓		✓	✓	✓				✓	
[62] Dimitropoulos et al.	2025	✓	✓	✓	✓	✓				✓	
[63] Wu et al.	2026	✓		✓	✓					✓	
[64] Alenjareghi et al.	2026		✓		✓			✓			
[65] Yin et al.	2026	✓			✓		✓			✓	
[66] Zhou et al.	2026	✓	✓	✓		✓	✓		✓		✓
[67] Liu et al.	2025	✓	✓	✓		✓	✓	✓			
[68] Gao et al.	2026	✓	✓	✓	✓				✓	✓	
[69] Hua et al.	2025	✓	✓		✓					✓	
[70] Liu et al.	2026	✓	✓	✓	✓	✓		✓			
[71] Yang et al.	2024	✓	✓		✓					✓	
[72] Mei et al.	2024	✓	✓	✓	✓					✓	
[73] Ding et al.	2023	✓			✓					✓	
[74] Lin et al.	2025	✓	✓	✓	✓	✓			✓		✓
[75] Zhao et al.	2023	✓	✓	✓	✓		✓		✓		✓
[76] Williams et al.	2025	✓	✓	✓	✓					✓	✓
[77] Chalvatzaki et al.	2023	✓			✓			✓		✓	
[78] Gorlo et al.	2024	✓	✓	✓	✓					✓	
[79] Zhang et al.	2024	✓	✓		✓		✓			✓	
[80] Oyama et al.	2024	✓	✓	✓	✓					✓	
[81] Liu et al.	2025	✓	✓	✓	✓					✓	
[82] Xia et al.	2025		✓			✓			✓		
[83] Wang et al.	2025		✓		✓		✓			✓	
[84] Song et al.	2025	✓			✓			✓		✓	
[85] Asuzu et al.	2025	✓	✓	✓	✓	✓				✓	
[86] Fan et al.	2024	✓	✓	✓	✓		✓		✓		
[87] Zheng et al.	2024	✓	✓	✓	✓		✓	✓			✓
[88] Liu et al.	2024	✓	✓	✓	✓					✓	
[89] Gao et al.	2024			✓	✓		✓			✓	
[90] Abbas et al.	2024			✓	✓	✓			✓		
[91] Graule et al.	2024	✓			✓					✓	
[92] Wanna et al.	2024		✓		✓	✓				✓	
[93] Joublin et al.	2024	✓			✓					✓	
[94] Izquierdo et al.	2024			✓	✓		✓		✓		
[95] Bu et al.	2025		✓		✓					✓	
[96] Mandi et al.	2024	✓	✓		✓					✓	
[97] Bashir et al.	2025	✓	✓		✓		✓		✓		
[98] Lin et al.	2025	✓	✓	✓	✓	✓				✓	
[99] Liu et al.	2024	✓	✓		✓				✓	✓	
[100] Song et al.	2024	✓	✓	✓	✓				✓	✓	
[101] Shi et al.	2025	✓	✓	✓	✓				✓	✓	
[102] Sun et al.	2025	✓	✓		✓			✓		✓	
[103] Wang et al.	2024	✓		✓	✓		✓			✓	
[104] Dorbala et al.	2024	✓	✓		✓	✓				✓	
[105] Lan et al.	2015	✓		✓	✓				✓		
[106] Tsushima et al.	2025	✓			✓					✓	
[107] Park et al.	2024	✓			✓	✓				✓	
[108] Ranasinghe et al.	2025	✓	✓		✓		✓			✓	
[109] Frering et al.	2025	✓	✓		✓					✓	
[110] Roy et al.	2025	✓	✓		✓	✓				✓	
[111] Li et al.	2025	✓	✓	✓	✓		✓			✓	
[112] Park et al.	2025	✓			✓	✓				✓	
[113] Gupta et al.	2024	✓			✓					✓	

AEA: Autonomous execution agents, ICA: Interactive coordination agents, MFA: Multimodal fusion agents
GCP: General corpus pre-training, DAP: domain-adaptive pre-training, SAP: safety-aligned pre-training
FPFT: Full parameter fine-tuning, PEFT: Parameter efficient fine-tuning, PFT: Prompt fine-tuning, RLFT: Reinforcement learning fine-tuning

communication paradigms. In comparison, HAI is largely focused on the core functions of information acquisition and exchange, leading to its typically more direct and simplified interaction modes. The safety paradigm for LLMs varies across domains. HRC demands the highest level of operational safety, requiring LLMs to internalize strict protocols to prevent physical hazards. Service robotics involves moderate physical safety concerns, while HAI prioritizes cybersecurity and data privacy as its primary safety objectives.

Through analysis, it is evident that the influence of LLMs in HRC is fundamental, facilitating a transition from automation to cognition. LLMs are required to evolve from being creative interlocutors or flexible service planners into production line engineers operating under stringent physical and socio-technical constraints. This shift necessitates not only an understanding of language but also a comprehension of, and adherence to, a complex system process standards, physical laws, and safety regulations. The ultimate pursuit of precision, reliability, certainty, and absolute accountability for physical consequences characterizes the distinctive role of LLMs in manufacturing applications. Furthermore, this specificity dictates that their technological development pathways must be independently designed rather than directly adopting paradigms from other fields.

LLMs agents for collaborative manufacturing

The advancement of LLMs agents is prompting HRC to transition from a reactive interaction model based on fixed rules to a collaborative partnership that emphasizes semantic understanding, task reasoning, and dynamic adaptation. To delineate the technical framework of LLMs agents within industrial HRC, we have classified them into three fundamental paradigms according to their functions and modes of interaction in collaborative systems, namely, autonomous agent, interaction coordination agent, and multi-modal fusion agent. This classification not only arises from an analysis of task structures, interaction frequencies, and perceptual complexities across various manufacturing scenarios but also reflects the distinct roles that LLMs assume in the process of translating high-level human instructions into robot-executable tasks.

Autonomous execution agents (AEA) focus on achieving high levels of automation in the execution of complex task sequences in structured environments [114,115]. AEA are concentrated in the lower right quadrant of Fig. 6, covering scenarios with high structured and low interaction requirements such as stamping/welding/spraying, large-scale assembly, batch quality inspection, and warehouse material sorting [63,65]. The common features of these scenarios are highly

standardized operation processes, strong controllability of the environment, and clear task goals [61]. The agents are well-suited for replacing humans in the execution of repetitive and high-precision physical tasks. They enhance production efficiency through closed-loop automation, necessitating human intervention only during task initiation and result acceptance [57]. However, the adaptability in dynamic environments is limited, and the transparency of decision-making processes remains low. Moreover, the absence of an effective human-robot handover mechanism under abnormal working conditions restricts their reliability in unstructured scenarios. Interactive coordination agents (ICA) are dedicated to achieving task collaboration and intention alignment between humans and robots in manufacturing environments [116,117]. As a dialogue partner on the production line, this type of agent excels at understanding compound instructions, managing multi-round dialogue states, and supporting joint decision-making [58,59]. The interactive coordination agents are densely concentrated in the upper left quadrant of Fig. 6, covering scenarios with low structure and high interaction requirements such as production line fault diagnosis, process parameter optimization, production plan adjustment, and quality root cause analysis [60,62,64]. These knowledge-intensive tasks have no fixed solution path and require in-depth dialogue and collaborative reasoning between the agent and domain experts. It transforms the empirical knowledge of human experts into executable operation plans through natural language interaction. The challenge of the agents lies in the length limitation of dialogue memory and the dynamic update of knowledge in the manufacturing field. Multimodal fusion agents (MFA) integrate visual, language and force control and other sensor information to build an overall understanding of the physical environment, thereby achieving more dexterous and adaptive HRC [118,119]. In complex manufacturing scenarios, MFA can understand the environmental context and incomplete human instructions, achieving precise task implementation [66,67]. MFA are mainly distributed in the lower left area of Fig. 6, involving scenarios with low structuring degree and interaction frequency such as human-robot assembly, flexible wiring harness/cable cabling, complex product disassembly, and safe collaborative handling [68,69]. Agents and human workers closely cooperate in the physical space, bridging the gap between semantic instructions and physical operations. Although it enhances environmental understanding and collaborative compliance, it still has deficiencies in cross-modal data alignment, training costs, and real-time performance [70]. The three types of agents enhance the capabilities of different dimensions within the HRC manufacturing framework. Autonomous agents facilitate closed-loop automation in task execution, interactive agents establish bridges for communication and collaboration, and multimodal agents provide the system with a understanding of both the environment and human operators. Rather than replacing one another, these agents form complementary technological systems. This architecture of intelligent agent specialization and collaboration, grounded in scene characteristics, offers a clear technical pathway for constructing an efficient, flexible, and secure HRC manufacturing system.

Pre-training and fine-tuning for HRC

In HRC manufacturing systems, the training strategies for LLMs differ fundamentally from those employed in general domains. General LLMs are primarily trained on extensive text corpora with the aim of achieving a broad understanding of language. In contrast, HRC-specific models integrate multimodal data from manufacturing environments, encompassing equipment status, sensor signals, process parameters, and more [120,121]. The primary objective of these dedicated models is to establish an accurate mapping between linguistic instructions and physical actions while addressing the specific requirements for safety, real-time performance, and reliability inherent in the manufacturing industry.

Pre-training is aimed at developing the essential cognitive capabilities required for manufacturing contexts [122,123]. This phase

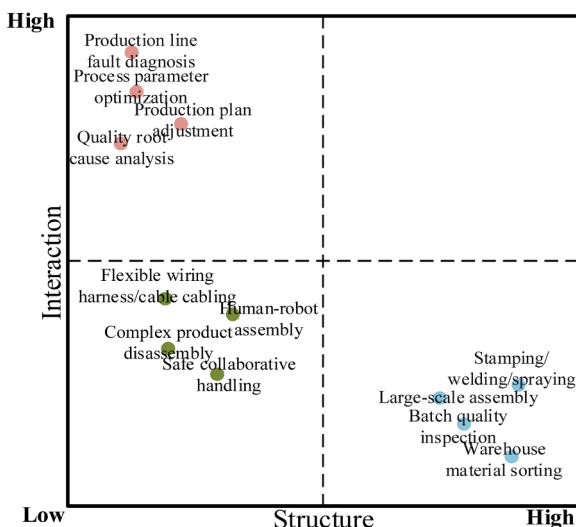


Fig. 6. Applicability of LLMs agents in HRC scenarios.

establishes a correlation mapping between linguistic instructions and physical operations through general corpus pre-training, thereby enabling the model to accurately interpret the specific physical meanings inherent in professional directives [58–60]. Concurrently, domain-adaptive pre-training constructs the model's manufacturing knowledge system by incorporating specialized information such as process specifications and equipment manuals [66,67]. Moreover, safety-aligned pre-training integrates industrial safety standards into the foundational logic of the model, ensuring that all ensuing decisions and actions adhere to the safety constraints relevant to the manufacturing environment [51,52]. Collectively, pre-training lays down the cognitive foundation of HRC, equipping it with a fundamental capacity to comprehend its operational surroundings and analyze intended actions effectively.

The fine-tuning strategy effectively adapts LLMs to specific manufacturing scenarios. In particular, in high-precision contexts such as aero engine assembly, full parameter fine-tuning allows the system to strictly comply with intricate assembly process requirements [124,125]. Conversely, in flexible manufacturing environments characterized by personalized customization, parameter-efficient fine-tuning demonstrates substantial advantages, reducing the production line adjustment cycle [126]. In guidance scenarios that involve frequent human-robot dialogue, prompt fine-tuning enhances the interaction template, enabling the system to accurately interpret informal expressions from operators and improve collaboration efficiency [127,128]. Furthermore, in safety-critical situations like hazardous material handling, reinforcement learning fine-tuning ensures that the system can make safe and reliable decisions within complex environments through mechanisms that incorporate human feedback [129,130].

The collaborative interaction between pre-training and fine-tuning is also evident in the construction of a continuous learning mechanism. In response to the ongoing emergence of new processes and equipment within the manufacturing environment, this mechanism ensures that the system assimilates new knowledge while preserving the stability of its original collaborative skills. Such a dynamic adaptation mechanism is crucial for HRC to sustain performance over extended periods of operation. By establishing foundational manufacturing cognition through pre-training and integrating it with scenario-based fine-tuning strategies, HRC have successfully achieved a precise transition from general intelligence to specialized capabilities. This hierarchical and progressive technical approach not only effectively addresses the personalized challenges inherent in the manufacturing environment but also offers a reliable solution for large-scale deployment of intelligent collaboration systems.

The cognitive hierarchy of LLMs-powered HRC

Based on the natural process and value creation chain of LLMs in solving problems in real manufacturing scenarios, we divide the application of LLMs in HRC into environmental perception and understanding, task planning and decision making, execution and interaction. It follows the natural progressive relationship of cognitive collaboration, forming a spiral upward path of ability evolution. This classification not only clearly outlines the capability boundaries of LLMs in HRC, but also demonstrates how they promote the transformation of HRC from instrumental assistance to cognitive collaboration through a progressive empowerment mechanism. An examination of the insights contained within the 65 key literature is presented in Table 4. Integrating findings across dimensions, this chapter conclusively answers RQ2.

Environmental perception and understanding

Effectively human-robot collaboration in manufacturing environments fundamentally relies on accurate perception of situational contexts. This chapter examines how LLMs process multimodal information to construct environmental awareness, as Fig. 7, progressing from

holistic multimodal context comprehension through precise interpretation of linguistic instructions to coordinated visuo-linguistic tracking. The sequential organization of these cognitive processes, from macroscopic scene understanding to microscopic object localization, establishes a robust foundation of environmental representation essential for subsequent decision-making stages.

Multimodal contextual awareness

Within industrial human-robot collaboration, achieving a cohesive and timely comprehension of the shared workspace is paramount. This capability, defined as multimodal contextual awareness (MCA), represents the synthesis of diverse information streams, including interpreted language commands, visually anchored object data, and physical feedback from sensors such as force and torque units [131,132]. MCA functions as a critical cognitive interface, translating multimodal inputs into a coherent representation of the environment's physical state, the ongoing task's progress, and the human operator's probable intentions [133]. This situational model is indispensable for informing subsequent stages of task planning and robotic execution [134].

Constructing reliable MCA systems presents specific difficulties inherent to industrial contexts. A central issue involves the accurate synchronization of heterogeneous, asynchronous data sources that are frequently corrupted by noise, all while operating under strict real-time deadlines, a set of constraints typically more severe than those in general AI domains [135]. Although LLMs are increasingly employed as central reasoning units for integration, a fundamental conflict can arise. Their strength in semantic reasoning and contextual linking may not align with the precise, safety-governed, and highly specialized demands of factory-floor tasks. Potential failures include the misinterpretation of vague instructions or an inability to correctly associate a linguistic descriptor with its corresponding physical entity, which can result in erroneous actions.

Current research illustrates both the prospects and constraints of existing MCA methodologies. For example, Graule et al. [91] show that LLMs, when provided with a semantic map of the surroundings, can anticipate human motions to enable proactive and safer robot adaptations. Likewise, systems like PlanCollabNL [94] employ LLMs to fluidly create and modify collaborative plans by merging high-level objectives with perceived scene conditions. Nevertheless, these approaches frequently depend on comparatively ideal sensory data and encounter challenges when dealing with the routine occlusions, disorder, and sensor inaccuracies found in actual industrial settings [92,93]. Consequently, evolving MCA necessitates a dedicated progression towards interaction-focused representation learning leveraging real-world robotic interaction data, flexible fusion mechanisms that evaluate and prioritize sensor confidence dynamically, and the creation of transparent reasoning models to ensure trustworthiness. Enhancing this integrative perceptual capability is vital for advancing HRC from constrained prototypes to robust, real-world manufacturing solutions.

Language instructions

Within industrial human-robot collaboration, language instructions (LIT) as a critical conduit for operational intent. This demands that LIT systems do more than understand general language. They must accurately interpret typically concise and domain-specific commands and reliably map them to precise actions within the dynamic physical and procedural context of the factory floor [136]. The underlying technology involves a continuous loop of parsing, decomposing, and executing instructions, a process that necessitates integrating structured manufacturing knowledge with the adaptive reasoning capabilities of LLMs [137].

A fundamental challenge in this domain stems from the tension between natural language flexibility and manufacturing's demand for deterministic execution. An instruction such as “inspect this weld” requires the system to visually resolve the reference, identify the correct inspection criteria from process documents, and position the robot

Table 4
Key literature review for HRC cognitive hierarchy.

Reference	Year	Task planning and decision making			Environmental perception and understanding			Embodied execution and interaction		
		RP	LHM	RR	LIT	VLT	MCA	NG	EQA	CA
[49] Lou et al.	2025	✓	✓	✓	✓			✓		
[50] Wang et al.	2026	✓	✓	✓				✓		
[51] Ma et al.	2026	✓	✓	✓	✓			✓		
[52] Chen et al.	2026	✓	✓	✓	✓	✓		✓	✓	✓
[53] Tasneem et al.	2026	✓	✓	✓	✓	✓		✓	✓	
[54] Wang et al.	2025	✓	✓	✓	✓					✓
[55] Yin et al.	2025		✓	✓	✓					✓
[56] Liu et al.	2026	✓	✓	✓	✓	✓		✓		✓
[57] Ding et al.	2026	✓	✓	✓	✓					✓
[58] Liu et al.	2026			✓	✓		✓	✓		✓
[59] Wang et al.	2026			✓	✓			✓		✓
[60] Wang et al.	2026			✓	✓			✓		✓
[61] Yu et al.	2025	✓		✓	✓					✓
[62]Dimitropoulos et al.	2025		✓	✓	✓					✓
[63] Wu et al.	2026			✓	✓					✓
[64] Alenjareghi et al.	2026			✓	✓				✓	✓
[65] Yin et al.	2026			✓		✓			✓	✓
[66] Zhou et al.	2026			✓						✓
[67] Liu et al.	2025		✓	✓			✓			✓
[68] Gao et al.	2026		✓	✓	✓					✓
[69] Hua et al.	2025		✓	✓	✓					✓
[70] Liu et al.	2026	✓	✓	✓	✓		✓	✓		✓
[71] Yang et al.	2024	✓	✓	✓	✓			✓		✓
[72] Mei et al.	2024	✓	✓	✓	✓	✓				✓
[73] Ding et al.	2023	✓	✓	✓	✓			✓		✓
[74] Lin et al.	2025	✓		✓	✓					✓
[75] Zhao et al.	2023	✓	✓	✓	✓					✓
[76] Williams et al.	2025	✓	✓	✓	✓					✓
[77] Chalvatzaki et al.	2023		✓	✓	✓			✓		✓
[78] Gorlo et al.	2024		✓	✓	✓			✓		✓
[79] Zhang et al.	2024		✓	✓	✓					✓
[80] Oyama et al.	2024			✓	✓	✓		✓		✓
[81] Liu et al.	2025	✓	✓	✓	✓	✓		✓		✓
[82] Xia et al.	2025		✓	✓	✓					✓
[83] Wang et al.	2025	✓	✓	✓	✓					✓
[84] Song et al.	2025			✓	✓					✓
[85] Asuzu et al.	2025	✓			✓	✓				✓
[86] Fan et al.	2024		✓	✓	✓	✓		✓		✓
[87] Zheng et al.	2024		✓	✓	✓	✓		✓		✓
[88] Liu et al.	2024		✓	✓	✓	✓		✓		✓
[89] Gao et al.	2024		✓	✓	✓	✓		✓		✓
[90] Abbas et al.	2024			✓	✓	✓		✓		✓
[91] Graule et al.	2024	✓	✓	✓	✓			✓	✓	✓
[92] Wanna et al.	2024	✓	✓	✓	✓			✓	✓	✓
[93] Joublin et al.	2024	✓	✓	✓	✓					✓
[94] Izquierdo et al.	2024		✓	✓	✓					✓
[95] Bu et al.	2025			✓	✓					✓
[96] Mandi et al.	2024	✓	✓	✓	✓					✓
[97] Bashir et al.	2025	✓	✓	✓	✓					✓
[98] Lin et al.	2025		✓	✓	✓		✓			✓
[99] Liu et al.	2024	✓	✓	✓	✓	✓		✓		✓
[100] Song et al.	2024	✓		✓	✓	✓		✓		✓
[101] Shi et al.	2025		✓		✓	✓		✓		✓
[102] Sun et al.	2025	✓	✓	✓	✓	✓		✓		✓
[103] Wang et al.	2024	✓	✓		✓	✓		✓		✓
[104] Dorbala et al.	2024	✓	✓	✓	✓	✓		✓		✓
[105] Lan et al.	2015	✓			✓	✓		✓	✓	✓
[106] Tsushima et al.	2025			✓	✓			✓	✓	✓
[107] Park et al.	2024			✓	✓				✓	✓
[108] Ranasinghe et al.	2025			✓	✓				✓	✓
[109] Frering et al.	2025			✓	✓			✓		✓
[110] Roy et al.	2025			✓						✓
[111] Li et al.	2025	✓	✓	✓						✓
[112] Park et al.	2025		✓	✓						✓
[113] Gupta et al.	2024	✓	✓	✓	✓					✓

RP: replanning, LHM: long-horizon manipulation, RR: representational reasoning;
LIT: language instructions, VLT: vision-language tracking, MCA: multimodal contextual awareness;
NG: navigation guidance, EQA: embodied question answering, CA: cognitive alignment.

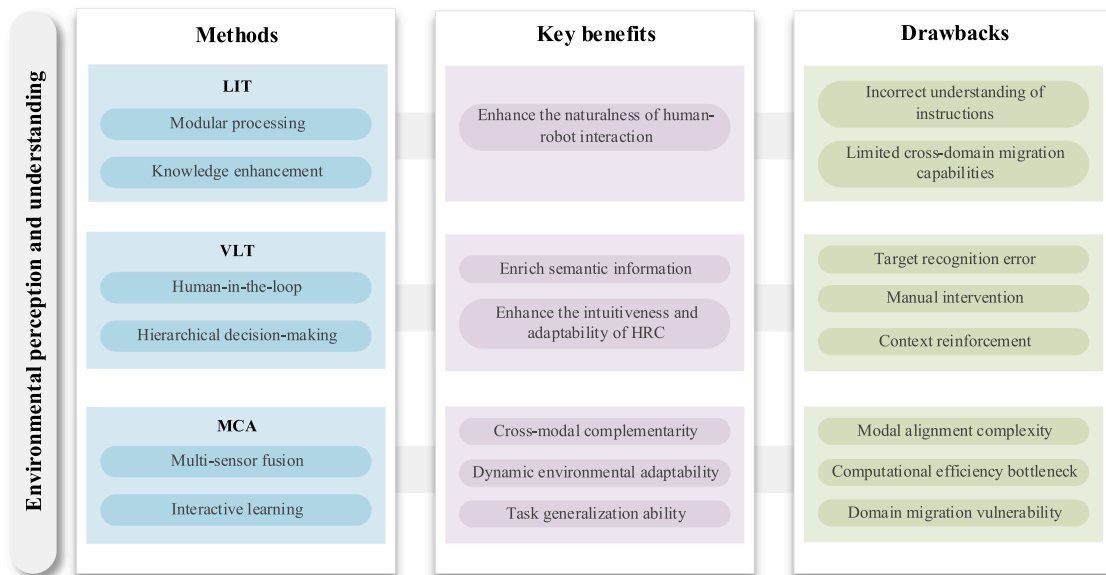


Fig. 7. Environmental perception and understanding methods of key benefits and challenges.

accordingly. Research efforts are creating specialized modules to address such needs. For example, Oyama et al. [80] integrated task classification and referential resolution with LLMs to improve the accuracy of instruction interpretation. Xia et al. [82] employed fine-tuned LLMs to reason about product lifecycle constraints for disassembly planning, directly tackling procedural uncertainty. Furthermore, Song et al. [84] combined linguistic understanding with structured knowledge graphs to enhance the robot's comprehension of physical object properties for more dexterous manipulation.

Despite these advances, significant barriers persist for real-world deployment. The inherent ambiguity of natural language poses a safety risk in precision-critical environments where a misinterpreted term can lead to faulty operations. Moreover, the knowledge grounding of general-purpose LLMs remains a concern, as they may generate plausible but physically incorrect or unsafe action sequences not aligned with certified protocols. Finally, the computational latency of sophisticated models often conflicts with the high-cycle-time requirements of production lines, creating a direct trade-off between reasoning depth and operational speed, a challenge also noted in distributed planning research [83]. Consequently, advancing LIT for industrial collaboration requires moving beyond generic language understanding. Future development must focus on deeply embedding domain-specific rules and safety constraints into the model's reasoning process, creating robust methods to verify the feasibility and safety of generated action plans, and engineering efficient, specialized models that meet the stringent throughput and reliability standards of industrial automation.

Vision-language tracking

Vision-language tracking (VLT) enables robotic systems to locate and follow objects based on natural language commands, a capability particularly valuable in dynamic manufacturing settings. By interpreting instructions like "track the moving conveyor belt" or "follow the orange wire harness," VLT reduces dependence on fixed programming and allows for more adaptive coordination [138]. This directly supports flexible task execution where visual targets or operational foci may change [139,140].

Central to this capability are visual language models (VLMs), which align visual inputs with textual queries in a unified representational space [141]. Applied to industrial collaboration, VLMs allow instructions to dynamically guide visual attention and tracking behavior. Research demonstrates this through various implementations. Some studies combine pre-trained language and vision models to achieve

dynamic task planning grounded in verbal commands [85]. Other frameworks integrate target retrieval with human-in-the-loop feedback to improve tracking precision during collaborative operations [86]. Further approaches leverage visual-language guidance within reinforcement learning to link object segmentation with subsequent task planning [87], while related work explores teleoperation interfaces and accelerated planning methods using similar principles [88,89].

However, the transition of VLT from controlled research to robust industrial application encounters significant hurdles. A primary challenge is the semantic-to-visual grounding ambiguity in complex workshops. Descriptions such as "the panel near the edge" can be misinterpreted under conditions of glare, occlusion, or cluttered backgrounds, leading to tracking failures. This ambiguity necessitates extensive, domain-specific training data that captures the visual diversity of real factory environments, data that is often costly and scarce. Moreover, achieving real-time inference performance is difficult, as fusing high-frame-rate video with continuous language processing demands substantial computational resources, potentially conflicting with the strict timing requirements of high-speed production lines. Future advancements must therefore prioritize industrial robustness. Key directions include developing VLMs trained on targeted, augmented datasets representing manufacturing edge cases, and designing efficient model architectures that ensure reliable, low-latency tracking performance amidst the visual noise and temporal constraints of real-world production.

Task planning and decision making

Building upon environmental perception, efficient task planning emerges as the central component of HRC. The present analysis delineates LLMs pivotal role in generating and adapting task plans, as Fig. 8, spanning three temporal dimensions of reasoning: dynamic replanning addresses immediate environmental fluctuations, long-horizon management orchestrates complex operational sequences, while representational reasoning supports logical analysis. This multi-layered architecture enables manufacturing systems to balance real-time responsiveness with strategic foresight in production planning.

Replanning

Manufacturing environments are inherently dynamic, where effective collaboration requires systems to adapt plans in real-time. Disruptions such as urgent order changes, equipment malfunctions, or

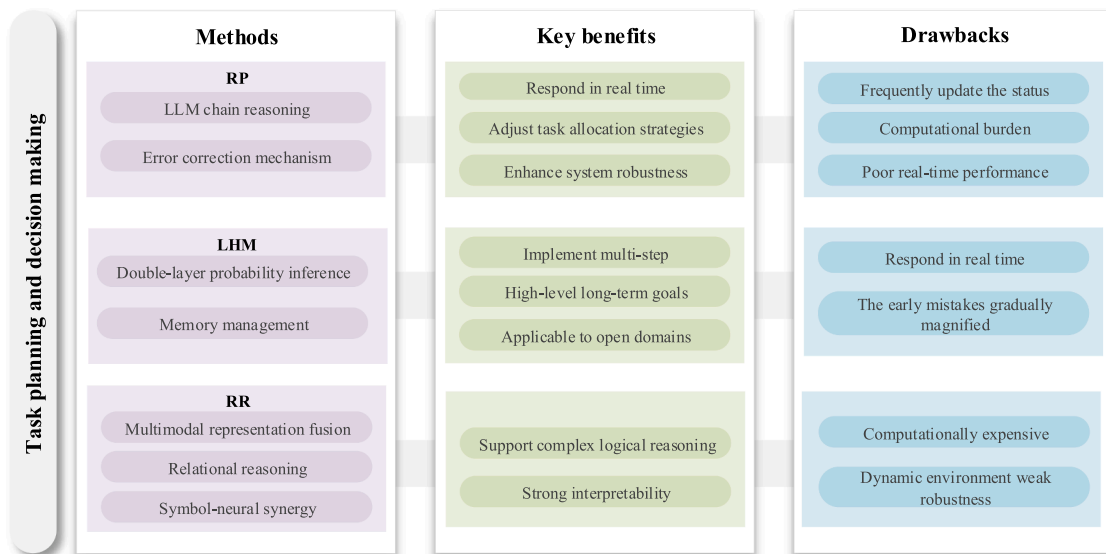


Fig. 8. Task planning and decision making methods of key benefits and challenges.

assembly errors necessitate swift and intelligent replanning to maintain production flow and safety [142,143]. This involves not only resequencing a robot's actions but also dynamically reallocating tasks between human and robotic agents. Challenge stems from the open and unpredictable nature of the shop floor, where changes in the task can introduce new objects or alter the scene layout, demanding that the system's world model be continuously updated [144].

The primary difficulty lies in generating new plans that are not only feasible but also safe, efficient, and executable within strict cycle-time constraints. Recent research investigates the use of LLMs to address this. For example, some frameworks utilize LLMs for chained reasoning to dynamically evolve task plans in response to verbal instructions or perceived changes [71]. Others have developed specialized VLMs that incorporate error correction mechanisms to detect and rectify planning mistakes based on visual feedback [72]. Furthermore, the common-sense reasoning of LLMs has been applied to semantically informed object rearrangement and motion planning, offering more context-aware adaptations [73].

However, these approaches often exhibit limitations when faced with the stringent demands of real-world production. A common shortcoming is the generation of static revised plans that lack true ongoing adaptability, failing to seamlessly incorporate live sensor feedback. This can result in plans that reference objects no longer present or that inadvertently lead to unsafe spatial interactions with human workers. The problem is especially pronounced in complex assembly tasks where conditions change rapidly. Advancing replanning for industrial use therefore requires methods that establish a tighter, real-time coupling between perception and planning. Future work must focus on developing algorithms that are deeply grounded in live sensor data, explicitly account for physical and safety constraints, and operate with sufficient speed to deliver reliable adaptations within the brief windows dictated by manufacturing process timing.

Long-horizon manipulation

Long-horizon manipulation (LHM) in industrial settings refers to complex, multi-step tasks such as assembling an engine component, wiring an electrical panel, or performing phased quality inspection. These tasks require a robot to execute a lengthy and logically ordered sequence of actions, often involving tool changes, interactions with multiple parts, and conditional branching based on intermediate outcomes [145,146]. Unlike simple pick-and-place, success in LHM demands not only precise low-level control but also high-level strategic planning and the ability to recover from or adapt to unexpected events,

as errors in early steps can propagate and compromise the entire operation [147].

LLMs offer a promising approach to managing this complexity by leveraging their capacity for commonsense reasoning and procedural understanding. Their ability to parse and generate sequential plans from language-based descriptions can reduce the need for exhaustive, manually programmed task specifications. For instance, research has shown that LLMs can enable robots to acquire and chain skills autonomously for unfamiliar long-cycle tasks, as demonstrated in work on self-driven skill acquisition frameworks [74]. Other approaches utilize LLMs within a two-tier probabilistic architecture to guide both high-level task planning and fine-grained motion inference, enhancing robustness [75]. Further developments integrate LLMs with force and visual feedback to dynamically adjust task execution in unpredictable situations [76], and employ language-structured memory networks to manage state across complex procedures [79].

Despite these advances, formidable challenges hinder the reliable deployment of LHM in real factories. A primary issue is the compounding of uncertainty and error over extended action sequences. An imperceptible misalignment in an initial assembly step may only cause a catastrophic failure many steps later, making diagnosis and recovery difficult. Furthermore, the long-term reasoning required often conflicts with the real-time performance demands of a production line. While LLMs may generate a plausible overall plan, ensuring that every constituent action is physically feasible, safe, and efficient within strict cycle times remains a significant hurdle. Therefore, advancing LHM for industrial use requires more than leveraging general planning capability. Future work must focus on developing verifiable and recoverable task sequences, where the system can monitor its own progress against explicit benchmarks and execute pre-defined or learned recovery routines. It also necessitates the tight integration of symbolic task planning with real-time, sensor-driven control loops to ensure plans are not only logically sound but also dynamically executable amidst the inherent noise and variability of the physical workshop.

Representation reasoning

Representation reasoning (RR) enables collaborative robotic systems to construct and manipulate an internal model of their tasks and surroundings. This goes beyond raw data processing to enable robots to understand relationships, infer unobserved states, and reason about cause and effect. For example, a system must know that “fastener A secures component B,” and that if A is missing, B will be unstable. This form of structured reasoning provides a crucial cognitive foundation for

interpretable decision-making in complex assembly or diagnosis tasks [148,149].

The application of this reasoning in manufacturing often involves fusing multimodal observations into a coherent representation, deducing relational knowledge between parts and tools, and combining learned neural patterns with explicit symbolic rules [150–152]. Recent work explores using LLMs to enhance these capabilities. Bu et al. [95] developed an error-aware method where LLMs generate reasoning chains for visual localization, reducing reliance on fixed heuristics. Mandi et al. [96] created a benchmarking platform to evaluate how well LLMs can represent and reason about multi-robot collaboration scenarios described in plain text. In a more direct application, Bashir et al. [97] used LLMs to perform dynamic reasoning on real-time product descriptions to predict assembly states and generate corresponding robot instructions.

However, deploying such reasoning robustly on the factory floor faces distinct challenges. A primary issue is reasoning latency and grounding in dynamic contexts. While LLMs generate a logically sound plan, it might do so too slowly for real-time control or may base its reasoning on outdated or incorrect perceptual data. Furthermore, there is often a mismatch between the statistical correlations learned by models and the deterministic physical causality governing manufacturing processes. This can lead to plans that are semantically plausible but physically unfeasible or unsafe. Therefore, advancing representation reasoning for industrial use requires a focus on causal and physics-aware modeling. Future studies need to integrate real-time sensor feedback tightly into the reasoning loop, develop representations that explicitly encode mechanical constraints and process knowledge, and achieve a level of computational efficiency compatible with the pace of production. This will move reasoning from a high-level planning tool to an integrated component of real-time, reliable collaborative operation.

Embodied execution and interaction

The ultimate validation of planning efficacy resides in physical world implementation. This chapter addresses how LLMs guide robots toward safe and reliable collaboration in authentic manufacturing settings, as Fig. 9, developing capabilities through three interconnected dimensions: spatial navigation establishes fundamental mobility, embodied questioning enables interactive environmental exploration, and cognitive alignment ensures value congruence between human and machine. The integrated progression of these competencies guarantees the effective

translation of planning into secure physical interactions while maintaining operational integrity.

Navigation guidance

Navigation guidance (NG) in manufacturing involves directing mobile robots or robotic arms within shared workspaces, a task that has evolved from basic collision avoidance to a form of collaborative motion planning. This requires systems to interpret not only the physical layout but also the intentions and ongoing actions of human workers, facilitating safe and efficient co-movement. Advanced approaches now integrate spatial reasoning with multimodal perception to understand context, such as predicting a worker's path or interpreting verbal instructions like “bring this to station three” [153].

Effective guidance hinges on several interrelated capabilities. It requires building a spatially and semantically rich map that includes not only static obstacles but also dynamic elements like human positions and temporary work-in-progress. It also depends on accurately aligning human intent with machine action, often inferred from a combination of language, gesture, and task context [154,155]. Research demonstrates progress in these areas. For instance, visual AI methods have been developed to guide mobile robots in assembly tasks [99]. Other work focuses on agents that can explore environments and backtrack for object retrieval based on language commands [101]. Furthermore, reasoning over room-object relationships using LLMs has been shown to improve target navigation with multimodal inputs [102], and serialized navigation decisions can be made by leveraging the commonsense reasoning of LLMs combined with visual object detection [104].

Despite these developments, deploying robust navigation guidance on the factory floor presents significant hurdles. A primary challenge is maintaining real-time stability and safety amid the unpredictability of human movement and the potential for sensor fusion errors. An LLM might plan a logically sound path, but minor delays in processing or misaligned sensor data can lead to dangerous situations. Additionally, the generalization of learned navigation policies across different factory layouts or task types remains difficult, often requiring extensive re-training or adaptation. Therefore, future advancements must prioritize real-time verifiable safety by tighter coupling of planning with instantaneous sensor feedback, and enhance cross-domain adaptability through simulation-to-real transfer learning and modular, composable navigation skills that can be reconfigured for various manufacturing scenarios.

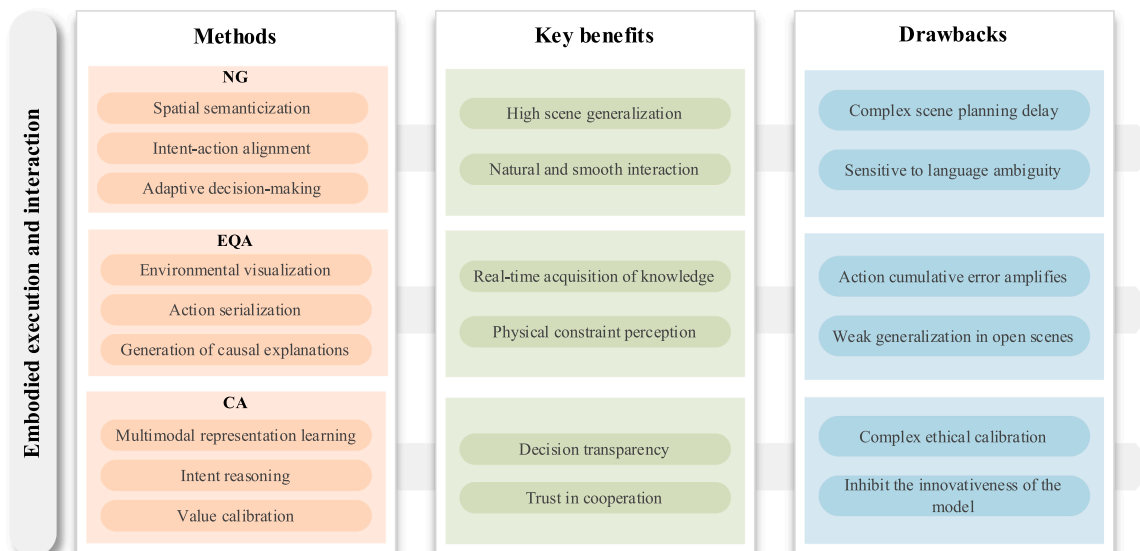


Fig. 9. Embodied execution and interaction methods of key benefits and challenges.

Embodied question answering

Embodied question answering (EQA) in industrial HRC enables a robot to answer queries about its physical environment through active exploration and interaction. Unlike retrieving information from a static database, EQA requires the robot to perceive, navigate, and manipulate its surroundings to gather evidence, such as checking an inventory bin or inspecting a part to answer “do we have enough M6 bolts?” or “is the weld seam complete?” [156,157]. This shift transforms the robot from a passive tool into an active investigative partner, capable of acquiring situational knowledge necessary for problem-solving [158]. The integration of LLMs has advanced EQA by enhancing environmental understanding and action planning. For example, Lan et al. [105] developed a structured semantic framework for EQA that enables iterative human-robot dialogue through layered visual perception. Tsushima et al. [106] created an LLM-based robotic assistant for two-way interactive task planning and adjustment via natural language. To ensure reliable interaction, Park et al. [107] proposed a method for evaluating whether user instructions given to such agents are clear, ambiguous, or unfeasible.

However, applying EQA in real manufacturing settings reveals significant technical gaps. The challenge is the semantic misalignment between language and perception. A question like “is the component properly aligned?” may be interpreted differently by the vision system depending on lighting or viewpoint, leading to incorrect answers. Furthermore, errors in the long sequence of actions needed to answer a question, such as moving to a location, identifying an object, and performing an inspection, can compound, reducing overall reliability. To advance EQA for practical use, researcher will focus on improving the robust grounding of language in the specific visual and physical context of the workshop. This requires more sophisticated multimodal alignment techniques and planning frameworks that are resilient to sensor noise and execution errors. Future systems should also incorporate explicit reasoning over domain knowledge and human feedback to correct mistakes and improve answer credibility, making EQA a trustworthy component of the collaborative workflow.

Cognitive alignment

Cognitive alignment (CA) is a fundamental requirement for effective and safe collaboration in industrial settings. It ensures that a robotic system’s internal decision-making logic, its interpretation of human intent, and its value judgments are compatible with those of its human partner. In practice, this means the robot must not only understand an instruction like “pass me the wrench” but also grasp the underlying goal, prioritize safety, and act in a predictable manner that aligns with the operator’s expectations and the factory’s protocols [159]. Achieving this requires integrating multimodal learning to perceive context, robust intention reasoning to infer goals, and mechanisms for value calibration to embed safety and efficiency norms into the agent’s choices [160,161].

Some studies are exploring various technical pathways to this alignment. For high-precision manufacturing scenarios, methods have been proposed to validate LLMs outputs against cognitive expectations to detect and mitigate context-induced errors or hallucinations [108]. Other approaches combine belief-desire-intention agent architectures with LLMs to generate semantic explanations for robot behavior, making its actions more interpretable to human workers [119]. Further work uses LLMs to generate anthropomorphic robot motions or select parameters that make robotic state transitions appear more natural and communicative [110]. Additionally, frameworks have been developed that enable robots to contextualize their actions dynamically based on the situation, improving adaptability in diverse service or assembly scenarios [113].

However, establishing reliable CA for industrial use faces profound challenges. A key issue is mitigating cognitive bias and value misalignment in dynamic environments. An LLM trained on general data may prioritize task completion speed in a way that conflicts with a site-specific safety culture emphasizing deliberate caution. Furthermore,

excessive or rigid alignment could stifle a system’s ability to propose novel, potentially more efficient solutions, effectively limiting its utility as a collaborative partner. Therefore, future research must move beyond seeking general compatibility. It needs to develop quantifiable, context-aware alignment metrics rooted in the cognitive demands of specific manufacturing tasks. This involves creating dynamic alignment mechanisms that can learn and adapt to individual operator styles or team norms without compromising core safety principles, enabling collaboration that is both trustworthy and genuinely synergistic.

Discussion challenges and future trends

The preceding passage extensively elucidated the deployment of LLMs within the HRC domain, illuminating their unparalleled capabilities in reasoning and generation. Nonetheless, as Fig. 10, beneath this veneer of promising innovation lies a multitude of significant challenges that urgently require resolution to safeguard the safety, efficacy, and ethical integrity of the HRC process. This chapter explores challenges and future research perspectives, answering RQ3.

Robustness and adaptability in dynamic environments

In real-world industrial manufacturing environments, the robustness and adaptability of HRC systems are confronted with substantial challenges. Most existing approaches rely on static datasets and predefined operational rules, which are inherently limited in their ability to accommodate real-time variations such as fluctuations in material properties, process adjustments, and dynamic human-robot interactions [162]. This limitation is particularly critical within the perception–decision–execution loop, where minor perceptual inaccuracies, such as errors in visual localization, can propagate and amplify through subsequent stages of task planning, ultimately resulting in assembly failures or deviations from specified tolerances [163]. For instance, in flexible assembly lines, current replanning strategies often depend on fixed rule sets or historical data. When confronted with unexpected variations in part poses or unforeseen process conflicts, these methods frequently incur computational delays, fail to satisfy stringent production cycle time requirements, and demonstrate inadequate fault tolerance [164,165]. The problem of error accumulation becomes even more acute in long-cycle manufacturing processes, such as engine assembly. Small deviations during initial operations can initiate a cascade of downstream disruptions, potentially leading to complete production line stoppages. Although reinforcement learning has shown promise in enhancing adaptability within dynamic environments, its reliance on sparse reward signals provides insufficient guidance for learning in complex, multi-step manufacturing tasks. This often results in slow system adaptation when encountering sudden changes in processes or shifts in collaborative dynamics between humans and robots [166].

To address these challenges, future research should closely align with industrial manufacturing scenarios by developing closed-loop systems that integrate world models with real-time feedback. World models should be capable of predicting changes in equipment status, material flow, and human behavior, while online simulation can reduce trial-and-error costs during system commissioning. Online few-shot learning techniques can decrease dependency on large-scale labeled datasets, enabling systems to adapt rapidly to new workpieces or processes through incremental updates. Furthermore, hybrid frameworks that combine symbolic reasoning with reinforcement learning could better balance exploration and exploitation in complex sequential tasks. Approaches such as hierarchical reinforcement learning and curriculum learning may also help alleviate sparse reward challenges in manufacturing settings, thereby enhancing long-term planning and real-time adjustment capabilities in dynamic production environments.

Semantic alignment and computational efficiency in multimodal fusion

Multimodal fusion is a critical enabler for advanced HRC. Within

	Existing Challenges	Future directions
Dynamic environments	• Relying on static data • Perceptual error • Sparse reward signal	• World Model • Online few-shot learning • Hierarchical reinforcement learning
Multimodal fusion	• Different modal semantic gap • Lack of interpretability constraints • Redundancy of high-dimensional feature mapping	• Neural-symbolic integration • Edge-cloud collaborative distributed inference • Embodied intelligent multimodal interaction
Ethical dilemmas	• Contextual illusions • Cognitive and value conflicts • Overalignment	• Quantitative alignment evaluation system • Dynamic optimization of model behavior • Multi-agent game mechanism • HRC authority and responsibility standards
Open-domain tasks	• Insufficient domain generalization ability • Agent skill migration • Limited to single-subject training	• Modular architecture • Causal reasoning • Common skill library & flexible task planner
Human-AI trust	• Semantic instruction ambiguity • Algorithm black-box • Decision opacity	• Bidirectional interpretability tools • Progressive trust models • Affective computing • Immersive collaborative environments

Fig. 10. Existing challenges and future directions.

industrial manufacturing, however, its practical implementation faces two persistent and intertwined challenges: cross-modal semantic alignment and the demand for computational efficiency under real-time constraints. In shop-floor collaboration, a significant semantic gap exists between different data modalities [167,168]. For instance, a human operator’s verbal instruction to “inspect this weld” must be correctly grounded in the pixel-level features captured by a vision system. Similarly, a textual process specification stating “fasten appropriately” lacks a direct quantitative mapping to real-time readings from a force-torque sensor. This semantic misalignment can lead to the robot misinterpreting intent, resulting in execution errors or safety risks. Current fusion architectures, often based on cascaded or parallel neural networks, introduce further limitations for industrial deployment [169]. Their interactions between modalities typically lack interpretable constraints based on physical or procedural knowledge. Furthermore, high-dimensional feature mappings create computational redundancy. The resulting processing delays are unacceptable in production environments where cycle times are critical and latency can directly cause line stoppages or hazardous situations.

Future research must therefore prioritize the development of lightweight and adaptive fusion architectures specifically designed for industrial settings. Neural-symbolic integration presents a promising direction. Incorporating symbolic rules, such as process knowledge, safety protocols, and physical constraints, can guide and constrain multimodal interaction. This approach not only enhances model interpretability and trustworthiness but can also effectively reduce parameter counts and computational overhead. Equally important is advancing distributed inference frameworks for edge-cloud collaboration. Offloading computation to edge devices near the data source minimizes transmission latency and is essential for meeting the stringent real-time requirements of production lines. Beyond traditional vision and language, integrating emerging modalities like haptic feedback offers new opportunities for industrial HRC [170,171]. The deep fusion of

force, vision, and language can better emulate a skilled worker’s perception-decision loop. This is particularly valuable for complex assembly or delicate manipulation tasks, ultimately leading to more natural, efficient, and robust collaborative operations.

Cognitive alignment and value safety in ethical dilemmas

The discordance between expansive models and human values has transcended mere technical considerations, emerging as a pivotal determinant in the sustainable evolution of HRC. Those systems exhibit a propensity to be deceived by contextual nuances when engaged in intricate tasks, culminating in what is referred to as an illusion, producing outputs that fail to align with the actual requisites of the task at hand [172]. This occurrence is especially pronounced in high-stakes precision endeavors, severely undermining the reliability and safety intrinsic to HRC [173]. Moreover, cultural disparities engender significant divergences in cognitive frameworks among diverse users. The collaborative practices and value systems that arise from varied cultural heritages may differ fundamentally, presenting formidable challenges for models operating within cross-cultural environments [174]. Nevertheless, it is imperative to recognize that excessive alignment may stifle the model’s innovative prowess, resulting in a detrimental loss of flexibility and adaptability.

To confront these ethical conundrums, it is imperative to develop a nuanced quantitative evaluation system for the future, such as alignment index. This would facilitate an objective measurement of the extent to which the model harmonizes with human intentions and values. By incorporating RLHF, the model’s behavior can be dynamically refined based on real-time human insights, thus enabling enhanced adaptability to the diverse needs of various users. Simultaneously, investigating value negotiation mechanisms within a multi-agent game framework is of paramount importance. Through interaction and negotiation among multiple agents, a delicate equilibrium can be

established between individual preferences and collective objectives, fostering a collaboration model that is both equitable and sensible. It is essential to establish standards delineating rights and responsibilities in HRC, ensuring transparency in responsibility attribution while creating tiers of clarity. These standards will not only furnish guidance for technological advancement but also safeguard users' rights and interests in tangible applications. The dual assurance provided by technology alongside policy frameworks holds significant potential to effectively navigate the ethical dilemmas associated with cognitive alignment and value security while steering HRC towards safer and more reliable avenues.

Scalability and generalizability in open-domain tasks

Although the HRC exhibits commendable performance in select tasks, its scalability and universality remain markedly constrained in the open-domain applications. The deficiency in domain generalization capabilities poses a fundamental limitation confronting the HRC framework [175]. In unfamiliar scenarios, the main reason for the performance decline of the existing system is the disconnection between task representation and potential causal mechanisms, coupled with the mismatch of the ontology during the skill transfer process [176]. A pivotal challenge lies in the scarcity of large-scale, diverse robotic datasets that can support the training of generalizable models. The recent emergence of large-scale open-source initiatives, such as the Open X-Embodiment project which consolidates data from multiple institutions and introduces the RT-X model family, marks a significant step towards building foundational models for robotics [177]. However, they are hindered by superficial feature alignment, making it arduous to capture invariant principles across diverse domains.

The intricacies associated with skill transfer represent a pivotal challenge that hampers the potential for universality. The inherent disparities among diverse robots and agents, manifesting in variations such as joint degrees of freedom and sensor configurations, render the transference of skills between different entities exceedingly complex [178]. Contemporary models frequently find themselves confined to training and optimization on a singular subject, thereby impeding the realization of the ideal objective: one-time learning applicable across multiple scenarios [179]. This limitation not only constrains the model's scalability, but also escalates both development and deployment costs.

To transcend these constraints, forthcoming research will pivot towards modular architectures. Such an architecture deconstructs intricate tasks into a series of relatively autonomous modules, exemplified by loosely coupled skill libraries and task planners. The skill library is designed to house a diverse array of ubiquitous skills, while the task planner adeptly activates the corresponding skill modules in accordance with specific task exigencies. This architectural paradigm not only facilitates plug-and-play functionality but also amplifies the flexibility and scalability of the model. Simultaneously, methodologies centered on causal reasoning are poised to emerge as a focal point of research. Causal reasoning possesses the capability to unearth transferable causal mechanisms from data correlations, thereby bolstering the model's adaptability across various scenarios.

Human-AI trust and collaborative paradigm innovation

The efficacy of HRC is inherently constrained by the naturalness of interaction and the mechanisms for building trust. A pivotal factor contributing to operational errors lies in the ambiguity surrounding semantic instructions. The inherent diversity and unpredictability of human language pose significant challenges for machines striving to accurately discern users' intentions, thereby undermining task execution [180]. Furthermore, skepticism towards opaque decision-making systems has attenuated the willingness to engage in collaborative endeavors. This distrust originates not only from the inscrutability of algorithms but also from a lack of transparency within their

decision-making processes. The absence of clarity fosters psychological resistance among users, while inflexible interaction paradigms fail to accommodate individual variances. Such asymmetries may engender considerable risks within scenarios requiring HRC. Equally paramount are innovations at the technological frontier. The advancement of affective computing alongside social signal recognition technologies promises to augment robots' capacity for empathy, enabling a more nuanced understanding of human emotions and needs [181]. By interpreting facial expressions and vocal intonations, robots can recalibrate their behaviors and feedback accordingly, thus enhancing both the naturalness and comfort level associated with collaboration. Moreover, introducing a mixed reality (MR) interface is set to transform user experiences by providing an immersive collaborative environment [182]. Allowing interaction with virtual agents through brain-computer interfaces (BCI), this engaging experience not only amplifies collaborative efficiency but also fortifies users' trust in these systems.

To cultivate a deeper mutual trust between humans and robots, it is imperative to develop sophisticated bidirectional interpretability tools in the foreseeable future [183]. A real-time decision traceability interface can eloquently present the model's decision-making process to users, offering an intuitive glimpse into how conclusions are drawn based on input information. Simultaneously, the introduction of a progressive trust model proves equally essential. By employing transparency classifications, we can enhance users' confidence in the system. Depending on the complexity and risk level inherent in various tasks, we should thoughtfully unveil the intricacies of the model's decision-making processes, enabling users to acclimate and foster genuine trust in the system's capabilities. The ultimate aspiration is to attain a symbiotic intelligence paradigm that not only prioritizes human leadership but also harnesses AI as a formidable enhancer. Within this framework, human wisdom and creativity will be fully unleashed while AI serves as a robust auxiliary tool, delivering efficient support and insightful counsel.

Conclusion

This work employed a systematic literature review methodology to curate and synthesize the research on human-robot collaboration powered by LLMs over the past three years. By conducting exhaustive searches across three prominent databases and adhering to rigorous selection criteria, a total of 1278 pivotal works were ultimately included for analysis. Utilizing bibliometric techniques, we undertook an in-depth examination of co-occurrence characteristics as well as evolutionary trends within thematic areas of this domain. The investigation centers on foundational knowledge surrounding LLMs and their potential applications within intelligent planning frameworks for HRC. The analysis is structured around three principal dimensions: environmental perception and understanding, task planning and decision making, execution and interaction, encompassing 9 subcategories. Our findings reveal that LLMs exhibit remarkable enhancements in flexibility and efficiency specifically in re-planning tasks within HRC contexts, long-term task orchestration, visual-linguistic tracking, and multimodal scene comprehension. These advancements provide essential support for collaborative endeavoring in dynamic environments. Concurrently, progress within embodied intelligence, encompassing representational reasoning, navigation guidance systems, interactive question answering mechanisms, and cognitive alignment, furnishes crucial insights that are instrumental in deepening our understanding of HRC dynamics.

Despite the aforementioned advancements, LLMs continue to grapple with pivotal challenges, such as inadequate real-time processing capabilities and limited adaptability to dynamic environments in HRC applications. To fully harness the potential of LLMs within human-centered intelligent manufacturing contexts, future inquiries should concentrate on investigating the following avenues: 1) Integrate a comprehensive world model with real-time feedback mechanisms to bolster environmental perception and responsiveness, 2) Develop a

lightweight adaptive fusion architecture aimed at enhancing deployment efficiency and operational efficacy, 3) Establish a quantitative evaluation framework for the system to scientifically appraise planning performance, 4) Design a modular architecture that augments both scalability and maintainability of the system, 5) Create bidirectional interpretability tools that enhance mutual trust between humans and robots, thereby fostering transparency in collaborative efforts. The breakthroughs achieved in these critical domains are anticipated to further deepen the integration and expand the application horizons of LLMs within actual human-machine collaboration systems, building upon existing accomplishments while driving development toward greater efficiency, robustness, and trustworthiness.

CRediT authorship contribution statement

Lihui Wang: Writing – review & editing, Formal analysis, Conceptualization. **Jinhua Xiao:** Writing – review & editing, Visualization, Validation, Formal analysis. **Gang Yuan:** Writing – original draft, Validation, Supervision, Software, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Maohua Xiao:** Validation, Resources, Methodology, Investigation, Data curation. **Xiaojun Liu:** Writing – review & editing, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing for financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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